

# Blockchain Basics - 3 Week Learning Plan

## Week 1: Foundations

- 1 Day 1: What is Blockchain? Understanding blocks, chains, and ledgers.
- 2 Day 2: Transactions, digital signatures, and how data is stored.
- 3 Day 3: Hashing and immutability.
- 4 Day 4: Public vs Private blockchain concepts.
- 5 Day 5: Nodes and how they maintain the ledger.
- 6 Day 6: Broadcast and peer-to-peer networking.
- 7 Day 7: Quick revision of the week.

## Week 2: Core Concepts & Ethereum Basics

- 1 Day 8: Decentralization – meaning and importance.
- 2 Day 9: Consensus mechanisms – introduction.
- 3 Day 10: Proof of Work (PoW) in detail.
- 4 Day 11: Proof of Stake (PoS) in detail.
- 5 Day 12: Mining and validation process.
- 6 Day 13: Ethereum vs Bitcoin – key differences.
- 7 Day 14: Smart Contracts – what they are and how they work.

## Week 3: Applications & Security

- 1 Day 15: Gas fee, wallets, and accounts in Ethereum.
- 2 Day 16: Blockchain transparency and immutability.
- 3 Day 17: Why blockchain is secure (cryptography, distribution).
- 4 Day 18: Double spending problem and its solution.
- 5 Day 19: Real-world applications – finance, supply chain, NFTs.
- 6 Day 20: Public vs Private keys (deep understanding).
- 7 Day 21: Full revision + prepare notes for Solidity.